



December 2025

(For IS)

Max. Marks: 100

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

- 1-

10. Which type of fiber cable provides minimum signal attenuation over the maximum distance, supporting runs up to 10 km?
 A) Standard connector (SC) B) Lucent connector (LC)
 C) Multi-mode fiber (MMF) D) Single-mode fiber (SMF)
11. In a Fibre Channel Arbitrated Loop (FC-AL) configuration, how many devices can perform I/O operations on the shared loop at any given time?
 A) One device B) Zero devices
 C) All devices simultaneously D) A maximum of 127 devices
12. Which Fibre Channel connectivity option involves two devices connected directly to each other, offering a dedicated connection but limited scalability?
 A) Mesh B) Point-to-point
 C) Arbitrated Loop D) Switched Fabric
13. In a Fibre Channel switched fabric (FC-SW), what is the term for the logical space in which all nodes communicate with one another?
 A) Domain B) Loop
 C) Zone D) Fabric
14. Which process is done at the array level, separate from zoning, to provide data access control by defining which Logical Unit Numbers (LUNs) a host can access?
 A) LUN Masking B) Zoning
 C) Storage Provisioning D) Port Zoning
15. In the context of Information Availability, what is the term for the period of time during which the system is not accessible?
 A) Timeliness window B) System downtime
 C) System dormancy D) System uptime
16. What metric is defined as the average time available for a system or component to perform its normal operations between failures?
 A) Recovery-Point Objective (RPO) B) Recovery-Time Objective (RTO)
 C) Mean Time Between Failure (MTBF) D) Mean Time To Repair (MTTR)
17. Which concept refers to the failure of a component that can terminate the availability of the entire system or IT service?
 A) Multipathing B) Fault Tolerance
 C) Server Clustering D) Single Point of Failure
18. In which backup type is a full backup performed initially, and subsequent backups only copy files that have been modified since the last incremental backup?
 A) Snapshot backup B) Incremental backup
 C) Differential backup D) Full backup
19. In a backup system's client/server architecture, which component manages the backup operations and maintains the backup catalog?
 A) Backup Server B) Backup Client
 C) Storage Node D) Application Host
20. Which security service provides the required secrecy of information and ensures that only authorized users have access to data, often through encryption?
 A) Accountability B) Availability
 C) Confidentiality D) Integrity

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain evolution of storage architecture with neat diagram.	8	L*2	1	1,2
	b) Explain Key characteristics of a data center with neat diagram.	8	L2	1	1,2
2.	a). Explain the types of RAID implementation methods.	8	L2	2	1,2
	b) Explain the concept of Mirroring in RAID with neat diagram.	8	L2	2	1,2

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|----|----|---|---|----|---|-----|
| 3. | a) | Explain the techniques involved in Cache Data Protection. | 8 | L2 | 2 | 1,2 |
| | b) | Explain traditional storage provisioning. | 8 | L2 | 2 | 1,2 |

Unit – II

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|----|----|---|---|----|---|-------|
| 4. | a) | Explain multimode fiber and single mode fiber cabling in FCSAN with neat diagram. | 8 | L2 | 3 | 1,2,3 |
| | b) | Explain the process involved in FC- AL transmission with a neat diagram. | 8 | L2 | 3 | 1,2,3 |
| 5. | a) | Explain Fiber Channel Protocol Stack with a neat diagram. | 8 | L2 | 3 | 1,2,3 |
| | b) | Explain single and multiple core topologies of Fiber Channel with neat diagram. | 8 | L2 | 3 | 1,2,3 |
| 6. | a) | Explain benefits of NAS. | 8 | L2 | 4 | 1,3 |
| | b) | Explain integrated NAS connectivity and gateway NAS connectivity. | 8 | L2 | 4 | 1,3 |

Unit – III

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|----|----|--|---|----|---|-----|
| 7. | a) | Explain implementation of fault tolerance in Business Community. | 8 | L2 | 5 | 1,2 |
| | b) | Explain back up methods. | 8 | L2 | 5 | 1,2 |
| 8. | a) | Explain the process involved in back up operation with a neat diagram. | 8 | L2 | 5 | 1,2 |
| | b) | Explain direct attached backup topology with a neat diagram. | 8 | L2 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
